# HW 9 – Tech Specifications

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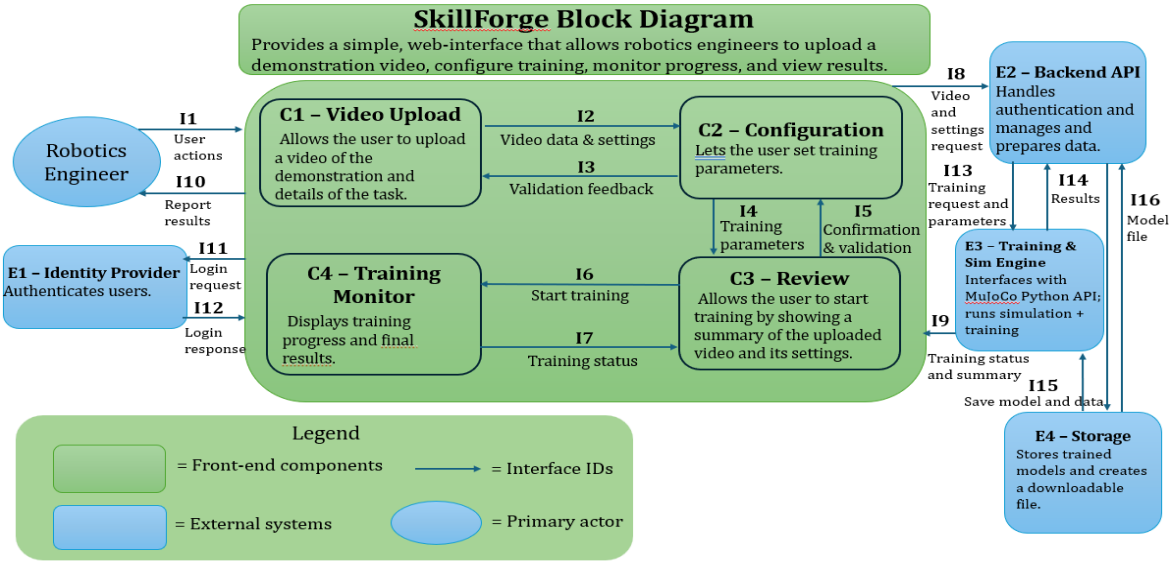
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## Version and Change Log

| | |
|---|---|
| V0.1 – Added architecture summary | 10/29/26 |
| V0.2 – Added tech stack section | 10/29/26 |
| V0.3 – Added interfaces and data flow | 10/30/26 |
| V0.4 – Added known constraints | 10/30/26 |
| V1.0 – Finalized traceability matrix | 10/30/26 |
| V1.1 – Updated Interfaces table and added summary endpoint grouping table | 10/30/26 |
| V1.2 – Added slightly updated D0 Arch design graphic | 10/30/26 |

# Architecture Summary:




**4 Major Components:**

## 1) Frontend (C1-C4)

The frontend system consists of components C1->C4 based on the design D0 document. The frontend comprises the systems for login, uploading videos for processing and training (C1), configuring run parameters (C2), and kicking off and monitoring training (C3&C4).

The frontend sends all client-stored values to the backend which are then processed. The frontend holds user identity info which is passed to the backend at each API call to verify identity. No communication is performed between the frontend and the database or training system.

The frontend stack is TypeScript + React, with other standards planned such as MOV for video file support.

## 2) Backend (E2 + E1)

The backend system is separated into 2 main functionalities. The first functionality is to support the frontend, primarily by providing an interface for secure communication with the database (storing videos, accessing created 3D data structures,…). The second functionality is to build and run the pipeline for simulation development and training runs.

The backend sends and receives data from the frontend (C1-C4), and communicates with the database (E4) to store and retrieve needed data. The backend communicates with the training and physics simulation component to run training and report back progress to the frontend. Potential connection with cloud GPU providers will be provided through the backend.

The backend will be written in Python with FastAPI apis.

**3) External Training & Physics Simulation System (E3)**

E3 comprises the backend components which do not have direct connection to the frontend. E3 accepts info about training video and developed meshes and auto-generated training annotations, and interfaces with MuJoCo APIs to develop the training data. Similarly, E3 supports training robot diffusion policies on the developed data.

E3 will need to interact with the database to pull relevant curated simulation and training data.

E3 will be written in Python with FastAPI apis.

**4) Database (E4)**

E4 is the database component and all scripting and support infrastructure required. E4 accepts and sends data to the backend components (E2 and E3) from SQL queries. Additionally, some scripting support like backup management may become part of E4 as well.

For initial test deployments, E4 will be written using SQLite as a fast and simple supported database. The file-based nature of SQLite can also help with reproducibility across multiple machines. For a more mature product alternative databases like Postgres will be investigated.

For the most part, a client-server framework will be used, with some backend components using a functional pipeline architecture.

# Interfaces and Data Formats

The API contract lists the endpoint, HTTP method, request, and response for each route. Other inter-component boundaries appear in the supplementary table below. Public API routes use /api/v1. JSON carries metadata and multipart/form-data carries files.

**Core Connecting API Endpoints**

| Endpoint | Method | Request | Response |
|---|---|---|---|
| /api/v1/auth/login | GET | Optional local return_to path (max 256 chars) | 302 redirect to OIDC provider |
| /api/v1/auth/callback | GET | Required code and state query strings | 302 to application; opaque session cookie |

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|---|---|---|---|
| /api/v1/auth/me | GET | Authenticated session cookie | 200 {authenticated, user_id} |
| /api/v1/auth/logout | POST | Session cookie, if present | 204 No Content; session cookie cleared |
| /api/v1/videos | POST | multipart/form-data: MP4/MOV video; size 1 byte to 2 GiB (proposed) | 201 {video_id, storage_key, size_bytes, sha256} |
| /api/v1/videos/{video_id} | GET | Path: positive integer video_id | 200 {video_id, storage_key, size_bytes, duration_s} |
| /api/v1/training-jobs | POST | JSON {video_id, parameters:{epochs, learning_rate, seed?}} | 201 {training_id, video_id, status, progress_pct} |
| /api/v1/training-jobs/{training_id} | GET | Path: positive integer training_id | 200 {training_id, video_id, parameters, status, progress_pct} |
| /api/v1/training-jobs/{training_id} | PATCH | Authorized worker: JSON {status, progress_pct}; version check proposed | 200 {training_id, status, progress_pct} |
| /api/v1/training-jobs/{training_id}/artifacts | POST | Authorized worker: multipart model binary + model_type | 201 {model_id, training_id, storage_key, size_bytes} |
| /api/v1/videos/{video_id}/point-clouds | POST | Authorized worker: multipart PLY/NPZ file | 201 {cache_id, video_id, storage_key, size_bytes} |

**Errors and Boundary Behavior**

| Endpoint group | Error response and recovery |
|---|---|
| Auth endpoints | 400 invalid redirect/code/state; 401 invalid/expired identity or session; 503 identity service unavailable; logout may return 500 on revocation failure |
| POST /videos | 400 empty/unsupported format; 409 duplicate checksum only if deduplication is enabled; 413 over approved limit; 500 storage failure; remove partial upload |
| GET /videos/{id} | 400 invalid ID; 404 unknown video; 500 metadata access error |
| POST /training-jobs | 404 video not found; 422 invalid training parameters; 500 database failure |
| GET /training-jobs/{id} | 400 invalid ID; 404 job not found; 500 database failure |
| PATCH /training-jobs/{id} | 404 unknown job; 409 stale update or invalid transition; 422 bad status/progress; 500 database failure |
| Artifact and point-cloud uploads | 400 unsupported point-cloud container; 404 parent job/video not found; 413 file limit; 422 missing model type; 500 storage failure; rollback and clean up |

JSON errors use {"error":{"code":"...","message":"..."}}. On failed uploads, delete incomplete files and roll back SQLite metadata. Access control is required for worker-only routes. Do not send internal file paths or credentials in responses

**Data Contracts and Formats**

Videos: MP4/MOV. D1/D2 proposed 1 byte–2 GiB uploads (not approved); D2 mentions >10 GB as a frontend edge case. Establish one agreed upload limit.

D1 metadata: VideoAsset relates one-to-many to TrainingJob and PointCloudCache; TrainingJob to ModelArtifact. Positive integer IDs, file keys, size in bytes, format and SHA-256 checksums; binary media files stored outside SQLite.

Training request JSON:
{"video_id":12,"parameters":{"epochs":100,"learning_rate":0.001,"seed":42}}. D1 ranges (provisional): epochs 1–1000; learning_rate >0 to 1; seed 0–2147483647.

Job status JSON: {"training_id":35,"video_id":12,"status":"queued","progress_pct":0.0}. Allowed states: queued/running/completed/failed; progress_pct numeric 0–100; stale updates rejected.

Artifacts: PLY/NPZ point cloud; PT/ONNX trained model. Exact mesh formats are tool dependent. Authentication: proposed OIDC with secure, opaque session cookie; provider unselected.

Open design decisions: D0 I15/I16 FTPS vs D1/D2 SQLite/filesystem; D0 I9 direct E3-to-C4 vs later E3-to-E2-to-C4; internal E2/E3 transport TBD

## Technology Stack

| Layer | Choice | Why |
|---|---|---|
| Frontend | React + Typescript | Team has prior experience; good open-source tools for things like streaming video upload UI components. |
| Backend | Python + pydantic + FastAPI | Team has prior experience + almost all ML tools provided through Python. Pydantic integration w/ FastAPI allows for input type checking and easy user token validation. |
| Database | SQLite | Fast, easy tools to use w/ Python backend. File-based storage allows for sharing database across devices without more extensive cross-device data sharing. |
| Hosting/Compute | Modal.ai | Free 35$/month tier, easy Python configuration |
| 3D-Photogametry | COLMAP | Industry standard for 3DGS. Easy to run from python files either with Python library bindings or through subprocess module. Integration with other libraries like fvdb is common. |
| 3DGS | Sparse-GS | Provides 3DGS capability when using a small number of unique views-even without something like a DepthAnything model as a prior. |
| Physics Simulation | MuJoCo | Faster processing speed vs IsaacSim + strong community support. |
| 3D Mesh Development | DNLR + Fixer | We chose the DNLR foundation model over basic TSDF fusion algorithm because DNLR offers higher visual fidelity. We accepted the trade-off of slower computation because high-quality meshes are 100% required in our case." |

## Known Constraints

These CON IDs are proposed from the team's Week 3 Constraints Essay. They identify inherited limitations and the concrete architectural consequence of each. Once the group approves IDs, reuse them unchanged in the Requirements Document and the Assignment 9 traceability matrix.

| **CON ID** | **Inherited limitation (Week 3)** | **Design consequence / exclusion** |
|---|---|---|
| CON-01 | Limited funds for GPU/CPU compute, storage, and streaming. Week 3 anticipates up to $150/month in Modal credits (~30$ in free tier credits per person) | Requires time-bounded GPU execution and monitoring of resource use; rules out assuming unlimited or always-on cloud training. Confirm provider credit amount and actual budget. |
| CON-02 | Training-video sharing is limited to subsections under 25 GB using student OneDrive accounts (Week 3). | Forces partitioning of shared training data; rules out assuming an unrestricted, centralized shared corpus. Different from the provisional 2-GiB per-file API upload cap. |
| CON-03 | 16GB Max VRAM (~4080ti size) | Forces profiling and a team-approved GPU-memory and cost ceiling before final NFR/test thresholds. Do not invent a current cap. |
| CON-04 | Team expertise in robotics and original neural-network research is limited (Week 3). | Favors reuse of established reconstruction, simulation, and ML libraries and advisor consultation; rules out making a novel robotics algorithm a required MVP dependency. |
| CON-05 | Long-term product vision may handle proprietary customer videos and payments (Week 3). | Forces restricted access and authenticated metadata/media routes. Payment processing is not established in the current prototype interface; if added later, payment details must not be stored after the transaction. |

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|---|---|---|
| CON-06 | The Week 3 security essay requires ingested training data to be fully deleted after use. | Forces a retention/deletion policy, cleanup of files and linked metadata, and tests for orphaned data. The team must define “after use” and whether trained artifacts are retained. |
| CON-07 | Isolation of proprietary data may cost more than shared low-cost compute (Week 3). | Requires documented choice of compute/storage isolation and cost trade-off before accepting proprietary input; rules out treating low-cost shared hosting as automatically sufficient. |

## Traceability Matrix

Aligned to Requirements Document, user stories, D0, D1, D2. Test IDs TC-001–TC-026 are proposed for Week 10. Status describes design documentation, not implementation completion. Confirm missing NFR thresholds and E4 data-deletion behavior before submission.

| **Req ID** | **Description** | **Design Component (D0/D1/D2)** | **Status** | **Test Case** |
|---|---|---|---|---|
| REQ-01 | Select and display demonstration video | D0 C1/I1–I2; D2 C1 video selection | Designed | TC-001 |
| REQ-02 | Validate uploaded video type and size | D0 C1/I1; D2 C1 validation; D1 E2 decoding | Designed | TC-002 |
| REQ-03 | Configure training parameters | D0 C2/I4–I5; D2 C2 parameter validation | Designed | TC-003 |
| REQ-04 | Review video and settings before processing | D0 C3/I4–I6; D2 C3 review | Designed | TC-004 |
| REQ-05 | Submit training request with parameters | D0 C3→E2/I8; D1 E3; D2 C3/E2 APIs | Designed | TC-005 |
| REQ-06 | Show training-run status | D0 C4/I7,I9; D1 E4 TrainingJob; D2 C4 | Designed | TC-006 |

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|---|---|---|---|---|
| REQ-07 | Show generated model artifact reference | D0 C4/E4/I14–I16; D1 E4 ModelArtifact; D2 C4 | Designed | TC-007 |
| REQ-08 | Display explained processing errors and codes | D0 I1–I16 error paths; D2 C1–C4/E2 | Designed | TC-008 |
| REQ-09 | Provide camera quality/viewpoint guidance | D1 E2 frame quality warnings; D2 C1 guidance | Planned | TC-009 |
| REQ-10 | Document model artifact format for downstream use | D0 E4/I15–I16; D1 E4 ModelArtifact; D2 E4 | Planned | TC-010 |
| NFR-01 | API response-time target (threshold missing in requirements) | D1/D2 E2 FastAPI routes and async job dispatch | Planned | TC-011 |
| NFR-02 | Require authentication for protected training operations | D0 E1/I11–I12; D2 E1 OIDC/PKCE and session validation | Designed | TC-012 |
| NFR-03 | Usability target (statement/threshold missing) | D0 C1–C4 workflow; D2 C1–C4 UI validation | Planned | TC-013 |
| NFR-04 | Persist training status through backend restarts (threshold missing) | D1 E4 TrainingJob; D2 E4 persistence; HW8 T24 | Designed | TC-014 |
| NFR-05 | Delete uploaded training data after use (details missing) | D1/D2 E4 VideoAsset/artifact lifecycle; HW8 T03 | Planned | TC-015 |
| CON-01 | Video storage in <25 GB subsections on student OneDrive | D0 E4 storage; D1/D2 E4 VideoAsset | Planned | TC-016 |
| CON-02 | Training/inference GPU memory ≤24 GB VRAM | D0 E3; D1/D2 E3 training and E2 reconstruction | Planned | TC-017 |
| CON-03 | Average GPU compute cost ≤$0.17/hour | D0 E3; D1/D2 E3 compute; HW8 T16–T20 | Planned | TC-018 |

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|---|---|---|---|---|
| CON-04 | Use available Modal AI credits up to $150/month | Technical Spec hosting/compute Modal.ai; D0 E3 | Planned | TC-019 |
| CON-05 | Scope limited by team robotics/ML expertise | D0 C1–C4/E2–E4; D1/D2 build-versus-reuse decisions | Designed | TC-020 |
| CON-06 | Seek advisor/professor guidance on expertise gaps | D1/D2 build-versus-reuse decisions; HW8 advisor milestone (process constraint) | Planned | TC-021 |
| CON-07 | Secure future customer payment processing | payment component excluded from current D0/D1/D2 release | Planned | TC-022 |
| CON-08 | Do not retain payment information after session | no payment/session interface in current D0/D1/D2 | Planned | TC-023 |
| CON-09 | Delete training data fully after use | D1/D2 E4 artifact/video lifecycle (deletion design unresolved) | Planned | TC-024 |
| CON-10 | Balance low-cost compute, security, deployment flexibility | D0 E1/E3/E4; D2 E1 security; Technical Spec hosting | Designed | TC-025 |
| CON-11 | Support portable deployment, e.g. Docker | D0 E2/E3/E4 client-server/pipeline; HW8 deployment/runbook T25 | Planned | TC-026 |

# Record of Advisor Review:

**Meeting Date: 5:30 PM 10/30/26**

**Document Suggestions:**

- Reformat interfaces and data formats to have same structure as D1 and D2 endpoints (Included in change V1.1)
- Include a visual diagram for the architecture summary (Included in V1.2)